

CARROLL, IA, USA

WMO: 725468

Lat: 42.046N

Lon: 94.789W

Elev: 367

StdP: 96.99

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04910

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.3	-18.6	-25.9	0.4	-20.2	-22.8	0.5	-17.9	15.2	-8.4	14.1	-5.9	3.9	290	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	33.7	24.8	32.4	24.4	30.9	23.5	26.9	31.5	25.8	30.2	24.6	29.1	6.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	22.2	30.6	24.1	19.9	28.9	22.9	18.4	27.7	87.2	31.2	82.1	30.1	76.6	29.7	35.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.6	10.4	-25.9	36.5	2.5	2.2	-27.7	38.0	-29.1	39.3	-30.5	40.6	-32.3	42.2		
(5)			-25.7	29.2	2.7	2.3	-27.7	30.9	-29.3	32.3	-30.8	33.6	-32.8	35.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	-5.7	-4.3	2.8	10.0	16.2	22.0	23.3	21.9	18.2	11.2	3.4	-3.7
	DBStd	11.80	6.95	7.19	6.86	5.54	4.99	3.69	3.15	3.11	4.70	5.66	6.01	6.52
	HDD10.0	1896	488	402	242	68	7	0	0	0	2	52	210	424
	HDD18.3	3659	746	635	484	256	102	9	3	6	59	230	448	683
	CDD10.0	1781	0	1	17	67	200	360	414	370	249	91	12	1
	CDD18.3	501	0	0	1	4	36	119	158	117	56	10	0	0
	CDH23.3	5155	0	0	8	91	412	1243	1594	1068	607	129	3	0
	CDH26.7	1892	0	0	1	25	143	494	616	358	216	39	0	0
Wind	WSAvg	4.6	5.3	5.2	5.3	5.8	5.3	4.3	3.1	2.7	3.4	4.5	5.1	4.9
Precipitation	PrecAvg	839	21	26	54	88	117	131	106	107	64	50	36	25
	PrecMax	1063	44	59	137	201	199	271	298	268	139	114	85	53
	PrecMin	621	3	4	1	25	37	30	33	14	14	6	0	0
	PrecStd	126	11	15	35	44	50	59	66	70	35	29	24	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	16.3	24.1	30.0	33.5	35.2	35.2	34.8	33.9	31.0	23.0	14.9
		MCWB	7.0	11.0	14.9	17.3	22.6	24.3	26.8	26.1	23.1	19.6	15.5	11.5
	2%	DB	7.9	12.2	20.2	26.1	30.2	33.0	32.9	32.4	31.9	27.2	19.0	10.2
		MCWB	4.7	8.7	13.5	16.5	20.6	23.8	25.9	25.7	22.6	18.0	13.4	7.4
	5%	DB	5.1	7.8	16.9	22.7	27.7	31.3	32.1	30.1	29.0	23.6	16.0	7.4
		MCWB	2.6	5.4	12.0	14.0	18.3	22.5	25.6	24.7	21.6	16.3	10.8	4.7
	10%	DB	2.7	4.9	12.9	20.0	25.1	29.8	30.1	28.0	27.3	21.0	12.7	4.9
		MCWB	1.2	2.9	8.8	12.8	17.2	21.6	24.6	23.2	20.2	15.0	8.4	2.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.7	12.1	17.4	19.1	24.4	26.9	28.8	28.2	25.7	23.0	16.6	12.3
		MCDB	11.3	15.9	23.2	26.6	30.9	32.3	33.6	32.9	30.8	28.0	21.4	14.7
	2%	WB	5.1	8.5	14.4	17.3	22.0	25.5	27.3	26.5	24.0	19.7	14.1	7.5
		MCDB	8.1	11.3	18.2	23.6	27.9	30.9	31.6	31.2	29.0	24.5	18.2	9.4
	5%	WB	2.9	5.5	12.0	15.4	20.1	24.1	26.2	25.2	22.8	17.7	11.3	4.8
		MCDB	5.1	8.2	16.6	21.6	25.4	29.3	30.4	29.3	27.4	21.6	15.0	7.2
	10%	WB	1.3	3.1	9.1	13.5	18.8	22.9	25.0	23.9	21.5	15.4	9.0	2.8
		MCDB	2.8	4.8	13.5	18.9	23.9	28.1	29.2	27.7	25.8	20.1	12.8	5.0
Mean Daily Temperature Range	5% DB	MDBR	9.2	9.4	10.7	12.8	12.2	11.7	11.4	11.6	13.7	12.8	11.0	8.8
		MCDBR	12.4	12.8	16.2	18.5	15.7	13.6	12.7	13.2	15.3	16.8	15.8	12.6
		MCWBR	9.7	9.8	10.3	10.6	7.9	6.7	7.5	8.4	8.3	9.4	10.1	9.9
	5% WB	MCDBR	11.4	12.6	14.9	16.1	13.3	11.8	11.3	11.8	12.8	13.7	14.4	11.9
		MCWBR	9.1	9.9	10.1	10.2	7.8	7.2	8.0	8.4	7.7	9.3	10.1	9.7
Clear-Sky Solar Irradiance	taub		0.282	0.299	0.320	0.373	0.398	0.415	0.427	0.414	0.381	0.336	0.295	0.280
	taud		2.421	2.386	2.419	2.293	2.295	2.296	2.246	2.288	2.337	2.453	2.500	2.451
	Ebn at Noon		871	909	929	895	878	860	845	845	849	855	852	842
	Edh at Noon		77	94	103	126	129	129	135	125	111	88	71	69
All-Sky Solar Radiation	RadAvg		1.93	2.95	3.80	4.79	5.66	6.28	6.44	5.50	4.53	3.09	2.11	1.59
	RadStd		0.18	0.21	0.36	0.40	0.42	0.53	0.41	0.38	0.31	0.38	0.15	0.13

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page